ALS Database Normalization - Updated Schema

NORMAL FORM STATUS		
Normal form	**Rule**	**Finding in the current schema**
1NF	Each field holds one value; repeating items are separate rows.	Meets: documents, enrollments, module progress, and game attempts are stored as individual records.
2NF	Each non-key value depends on the whole candidate key.	Meets: progress is unique per learner/module pair; game attempts are unique per learner/attempt UUID.
3NF	Non-key values do not depend on other non-key values.	Broadly meets: learner, center, curriculum, progress, and attempt facts are stored in related tables. Review the conditional items on page 2.

1NF | FIRST NORMAL FORM		
Record group	**One row represents**	**Current table or tables**
Account	One login account.	users
Learner profile	One profile for one account.	user_profiles
Enrollment	One program enrollment record.	enrollments
Uploaded document	One document and its review state.	learner_documents
Curriculum	One strand or one module.	learning_strands; modules
Reading progress	One learner’s progress for one module.	module_downloads_and_progress
Game activity	One game attempt.	module_game_attempts
Portal banner	One homepage banner.	hero_banners

2NF | SECOND NORMAL FORM		
Fact	**Key dependency**	**Current normalized table**
Login/account facts	user_id -> username, email, role, status	users
Demographic/profile facts	user_id -> profile fields	user_profiles; user_id is unique
Module facts	module_id -> module title, strand, level, language, file details	modules
Reading state	(user_id, module_id) -> progress fields	module_downloads_and_progress; pair is unique
Game attempt facts	(user_id, attempt_uuid) -> attempt fields	module_game_attempts; pair is unique

ALS Database Normalization - Updated Schema (continued)

3NF | THIRD NORMAL FORM		
Dependency / possible transitive fact	**How the current schema separates it**	
module_id -> strand_id; strand_id -> strand name, code, description	Strand details are in learning_strands; modules store strand_id.	
user_id -> account credentials and access role	Account facts are in users.	
user_id -> demographic and education profile	Profile facts are in user_profiles, one row per account.	
clc_id -> center name, code, location, status	Center details are in community_learning_centers; enrollments reference clc_id.	
module_id -> module metadata; learner/module -> progress	Module details are in modules; progress values are in module_downloads_and_progress.	

3NF REVIEW | BUSINESS RULES TO CONFIRM		
Area	**Current fields / overlap**	**Normalization check**
Address	user_profiles stores barangay, city, and province.	If barangay determines city/province, use location reference tables and store a location key. If location is fixed to San Juan, document that assumption.
Prior ALS history	user_profiles stores one previous program/year/completion set.	If a learner can have multiple prior periods, use one learner_education_history row per period.
Teacher assignment	A teacher can be assigned on both a CLC and an enrollment.	If enrollment always inherits the CLC teacher, avoid storing the same assignment twice. Retain both when enrollment assignment is an override or historical snapshot.

FINAL APPLICATION TABLES		
Data area	**Current tables**	
Accounts and profile	users; user_profiles	
Centers and intake	community_learning_centers; enrollments; learner_documents	
Learning content	learning_strands; modules	
Learning activity	module_downloads_and_progress; module_game_attempts	
Portal content	hero_banners	
Framework support	password_reset_tokens; sessions; cache; cache_locks; jobs; job_batches; failed_jobs; personal_access_tokens	

ALS Database Tables - Updated Schema

users		
Field	**Type / Key**	**Description**
id	BIGINT UNSIGNED | PK; AUTO_INCREMENT	Account identifier.
lrn	VARCHAR(12) | UQ; NULL	Learner Reference Number; unique when provided.
username	VARCHAR(255) | UQ	Unique username.
email	VARCHAR(255) | UQ; NULL	Email address; unique when provided.
email_verified_at	TIMESTAMP | NULL	Email verification time.
password	VARCHAR(255)	Hashed password.
role	ENUM('super_admin', 'admin', 'mobile_teacher', 'learner') | DEFAULT 'learner'	Account role.
status	ENUM('active', 'inactive') | DEFAULT 'active'	Account status.
last_login_at	TIMESTAMP | NULL	Most recent login time.
remember_token	VARCHAR(100) | NULL	Remember-me token.
created_at	TIMESTAMP | NULL	Record creation time.
updated_at	TIMESTAMP | NULL	Most recent update time.
deleted_at	TIMESTAMP | NULL	Soft-delete time.
role, status	INDEX	Composite index on account role and status.
lrn	INDEX	Additional index; the column is also unique.
username	INDEX	Additional index; the column is also unique.

user_profiles		
Field	**Type / Key**	**Description**
id	BIGINT UNSIGNED | PK; AUTO_INCREMENT	Profile identifier.
user_id	BIGINT UNSIGNED | FK; UQ	References users.id; cascade on delete; one profile per user.
first_name	VARCHAR(255)	First name.
middle_name	VARCHAR(255) | NULL	Middle name.
last_name	VARCHAR(255)	Last name.
name_extension	VARCHAR(10) | NULL	Name extension, such as Jr. or III.
birthdate	DATE	Date of birth.
gender	ENUM('male', 'female', 'other') | DEFAULT 'male'	Gender.
civil_status	ENUM('single', 'married', 'widowed', 'separated', 'solo_parent', 'other') | DEFAULT 'single'	Civil status.
religion	VARCHAR(255) | NULL	Religion.
contact_number	VARCHAR(25)	Primary contact number.
facebook_name	VARCHAR(255) | NULL	Facebook name.
street_address	VARCHAR(255)	Street or house address.
barangay	VARCHAR(255) | INDEX	Barangay.
city	VARCHAR(255) | DEFAULT 'San Juan City'	City.

user_profiles

Field	Type / Key	Description
province	VARCHAR(255) | DEFAULT 'Metro Manila'	Province.
highest_education_attained	VARCHAR(255)	Highest education attained.
dropout_reason	VARCHAR(255) | NULL	Reason for leaving prior schooling.
has_attended_als	BOOLEAN | DEFAULT FALSE	Whether the learner previously attended ALS.
previous_als_program	VARCHAR(255) | NULL	Previously attended ALS program.
previous_als_year	VARCHAR(20) | NULL	Year of previous ALS attendance.
previous_als_completed	BOOLEAN | NULL	Whether the previous ALS program was completed.
previous_als_incomplete_reason	VARCHAR(255) | NULL	Reason previous ALS program was incomplete.
mother_tongue	VARCHAR(255) | DEFAULT 'Tagalog / Filipino'	Mother tongue.
is_ip	BOOLEAN | DEFAULT FALSE	Indigenous Peoples indicator.
ip_community_name	VARCHAR(255) | NULL	Indigenous community name.
is_pwd	BOOLEAN | DEFAULT FALSE	Person with disability indicator.
pwd_disability_type	VARCHAR(255) | NULL	Disability type.
is_4ps_beneficiary	BOOLEAN | DEFAULT FALSE	4Ps beneficiary indicator.
father_name	VARCHAR(255) | NULL	Father's name.
mother_name	VARCHAR(255) | NULL	Mother's name.
guardian_name	VARCHAR(255) | NULL	Guardian's name.
guardian_relationship	VARCHAR(255) | NULL	Relationship to guardian.
emergency_contact_name	VARCHAR(255) | NULL	Emergency contact name.
emergency_contact_number	VARCHAR(25) | NULL	Emergency contact number.
profile_photo_path	VARCHAR(255) | NULL	Stored profile photo path.
created_at	TIMESTAMP | NULL	Record creation time.
updated_at	TIMESTAMP | NULL	Most recent update time.
deleted_at	TIMESTAMP | NULL	Soft-delete time.
last_name, first_name	INDEX	Composite index for profile-name queries.

community_learning_centers

Field	Type / Key	Description
id	BIGINT UNSIGNED | PK; AUTO_INCREMENT	Center identifier.
clc_name	VARCHAR(255)	Community Learning Center name.
clc_code	VARCHAR(30) | UQ	Unique center code.
clc_type	VARCHAR(255) | DEFAULT 'Community Center'	Center type.
barangay	VARCHAR(255) | INDEX	Barangay location.
physical_location_notes	TEXT | NULL	Directions or location notes.
assigned_teacher_id	BIGINT UNSIGNED | FK; NULL; INDEX	References users.id; set null if teacher is deleted.
status	ENUM('active', 'inactive') | DEFAULT 'active'; INDEX	Center status.
created_at	TIMESTAMP | NULL	Record creation time.
updated_at	TIMESTAMP | NULL	Most recent update time.
deleted_at	TIMESTAMP | NULL	Soft-delete time.

enrollments

Field	Type / Key	Description
id	BIGINT UNSIGNED | PK; AUTO_INCREMENT	Enrollment identifier.
user_id	BIGINT UNSIGNED | FK	References users.id; cascade on delete.
calendar_year	VARCHAR(10) | DEFAULT '2026'	Enrollment calendar year.
program_track	ENUM('BLP', 'A&E_Elementary', 'A&E_Junior_High')	ALS program track.
clc_id	BIGINT UNSIGNED | FK	References community_learning_centers.id; cascade on delete.
assigned_teacher_id	BIGINT UNSIGNED | FK; NULL	Optional users.id teacher reference; set null on delete.
enrollment_status	ENUM('pending', 'under_review', 'enrolled', 'needs_correction', 'rejected', 'withdrawn', 'completed') | DEFAULT 'pending'; INDEX	Current enrollment status.
delivery_mode	ENUM('modular_offline', 'modular_print', 'face_to_face', 'blended') | DEFAULT 'modular_offline'	Learning delivery mode.
schedule_preference	ENUM('weekend', 'weekday_am', 'weekday_pm', 'flexible') | DEFAULT 'flexible'	Preferred schedule.
remarks	TEXT | NULL	Enrollment notes.
enrolled_at	TIMESTAMP | NULL	Enrollment confirmation time.
completed_at	TIMESTAMP | NULL	Completion time.
created_at	TIMESTAMP | NULL	Record creation time.
updated_at	TIMESTAMP | NULL	Most recent update time.
deleted_at	TIMESTAMP | NULL	Soft-delete time.
user_id, calendar_year	INDEX	Composite index for learner/year queries.
clc_id, enrollment_status	INDEX	Composite center/status index.
enrollment_status	INDEX	Additional status index.
calendar_year	INDEX	Additional year index.
program_track	INDEX	Program-track index.

learner_documents

Field	Type / Key	Description
id	BIGINT UNSIGNED | PK; AUTO_INCREMENT	Document identifier.
user_id	BIGINT UNSIGNED | FK	References users.id; cascade on delete.
enrollment_id	BIGINT UNSIGNED | FK; NULL; INDEX	Optional enrollments.id reference; cascade on delete.
document_type	ENUM('psa_birth_cert', 'barangay_clearance', 'form_137_sf10', 'photo_id', 'portfolio_document')	Uploaded document category.
file_path	VARCHAR(255)	Stored file path.
file_name	VARCHAR(255)	File name.
file_size	BIGINT UNSIGNED	File size in bytes.
mime_type	VARCHAR(100)	File media type.

learner_documents

Field	Type / Key	Description
verification_status	ENUM('pending', 'verified', 'rejected') | DEFAULT 'pending'; INDEX	Document review status.
verification_notes	TEXT | NULL	Reviewer feedback.
reviewed_by	BIGINT UNSIGNED | FK; NULL	Optional users.id reviewer; set null on delete.
reviewed_at	TIMESTAMP | NULL	Review time.
created_at	TIMESTAMP | NULL	Record creation time.
updated_at	TIMESTAMP | NULL	Most recent update time.
deleted_at	TIMESTAMP | NULL	Soft-delete time.
user_id, document_type	INDEX	Composite learner/document-type index.

learning_strands

Field	Type / Key	Description
id	BIGINT UNSIGNED | PK; AUTO_INCREMENT	Learning strand identifier.
code	VARCHAR(20) | UQ; INDEX	Unique strand code.
name	VARCHAR(255)	Learning strand name.
description	TEXT | NULL	Strand description.
color_code	VARCHAR(30) | DEFAULT '#2563eb'	Display color.
icon_name	VARCHAR(50) | DEFAULT 'book-open'	Display icon key.
order_seq	INT UNSIGNED | DEFAULT 1; INDEX	Display order.
created_at	TIMESTAMP | NULL	Record creation time.
updated_at	TIMESTAMP | NULL	Most recent update time.

modules

Field	Type / Key	Description
id	BIGINT UNSIGNED | PK; AUTO_INCREMENT	Module identifier.
strand_id	BIGINT UNSIGNED | FK	References learning_strands.id; cascade on delete.
level	ENUM('BLP', 'Elementary', 'Junior_High')	Education level.
sub_level	VARCHAR(50) | NULL; INDEX	Sub-level, such as lower or advanced elementary.
language	VARCHAR(20) | DEFAULT 'English'; INDEX	Module language.
module_number	VARCHAR(50)	Module sequence or label.
title	VARCHAR(255)	Module title.
description	TEXT | NULL	Module summary.
pdf_file_path	VARCHAR(255)	Stored PDF path.
pdf_file_size	BIGINT UNSIGNED | DEFAULT 0	PDF size in bytes.
version	VARCHAR(20) | DEFAULT '1.0'	Module version.
is_published	BOOLEAN | DEFAULT TRUE; INDEX	Whether learners can access the module.
uploaded_by	BIGINT UNSIGNED | FK	References users.id; cascade on delete.
created_at	TIMESTAMP | NULL	Record creation time.
updated_at	TIMESTAMP | NULL	Most recent update time.
deleted_at	TIMESTAMP | NULL	Soft-delete time.
strand_id, level	INDEX	Composite strand/level index.

modules		
Field	**Type / Key**	**Description**
level, is_published	INDEX	Composite level/publication index.
level, sub_level, language	INDEX	Composite catalog filter index.

module_downloads_and_progress		
Field	**Type / Key**	**Description**
id	BIGINT UNSIGNED | PK; AUTO_INCREMENT	Progress record identifier.
user_id	BIGINT UNSIGNED | FK	References users.id; cascade on delete.
module_id	BIGINT UNSIGNED | FK	References modules.id; cascade on delete.
downloaded_at	TIMESTAMP | NULL	Most recent download time.
last_opened_at	TIMESTAMP | NULL	Most recent open time.
completion_status	ENUM('downloaded', 'in_progress', 'completed') | DEFAULT 'downloaded'; INDEX	Module completion state.
device_identifier	VARCHAR(255) | NULL	Mobile device identifier.
last_page_read	INT UNSIGNED | DEFAULT 1	Last page reached.
total_pages	INT UNSIGNED | NULL	Total PDF page count.
synced_at	TIMESTAMP | NULL; INDEX	Most recent synchronization time.
created_at	TIMESTAMP | NULL	Record creation time.
updated_at	TIMESTAMP | NULL	Most recent update time.
user_id, module_id	COMPOSITE UQ	One progress record per learner and module.

module_game_attempts		
Field	**Type / Key**	**Description**
id	BIGINT UNSIGNED | PK; AUTO_INCREMENT	Game attempt row identifier.
attempt_uuid	CHAR(36)	Client-generated attempt UUID.
user_id	BIGINT UNSIGNED | FK	References users.id; cascade on delete.
module_id	BIGINT UNSIGNED | FK	References modules.id; cascade on delete.
game_type	VARCHAR(50) | DEFAULT 'memory_match'	Game type.
score	SMALLINT UNSIGNED	Earned score.
max_score	SMALLINT UNSIGNED	Maximum possible score.
moves	SMALLINT UNSIGNED	Move count.
elapsed_seconds	INT UNSIGNED	Attempt duration in seconds.
device_identifier	VARCHAR(150) | NULL	Mobile device identifier.
attempted_at	TIMESTAMP	Attempt time.
created_at	TIMESTAMP | NULL	Record creation time.
updated_at	TIMESTAMP | NULL	Most recent update time.
user_id, attempt_uuid	COMPOSITE UQ	Prevents syncing the same attempt twice for one learner.
user_id, attempted_at	INDEX	Composite learner/time index.
module_id, attempted_at	INDEX	Composite module/time index.

hero_banners

Field	Type / Key	Description
id	BIGINT UNSIGNED | PK; AUTO_INCREMENT	Banner identifier.
title	VARCHAR(255) | NULL	Banner title.
subtitle	VARCHAR(255) | NULL	Banner subtitle.
image_path	VARCHAR(255)	Stored banner image path.
position_x	INT | DEFAULT 50	Horizontal image focal position.
position_y	INT | DEFAULT 50	Vertical image focal position.
zoom	SMALLINT UNSIGNED | DEFAULT 100	Image zoom percentage.
button_text	VARCHAR(255) | NULL; DEFAULT 'Take A Tour'	Call-to-action button text.
button_url	VARCHAR(255) | NULL; DEFAULT '/register'	Call-to-action destination.
order_seq	INT | DEFAULT 0	Display order.
is_active	BOOLEAN | DEFAULT TRUE	Whether the banner is active.
created_at	TIMESTAMP | NULL	Record creation time.
updated_at	TIMESTAMP | NULL	Most recent update time.

password_reset_tokens

Field	Type / Key	Description
email	VARCHAR(255) | PK	Account email used for password reset.
token	VARCHAR(255)	Password reset token.
created_at	TIMESTAMP | NULL	Token creation time.

sessions

Field	Type / Key	Description
id	VARCHAR(255) | PK	Session identifier.
user_id	BIGINT UNSIGNED | NULL; INDEX	Optional user reference; indexed, with no foreign key declared.
ip_address	VARCHAR(45) | NULL	Session IP address.
user_agent	TEXT | NULL	Browser or device user agent.
payload	LONGTEXT	Serialized session payload.
last_activity	INT | INDEX	Last activity time as a Unix timestamp.

cache

Field	Type / Key	Description
key	VARCHAR(255) | PK	Cache key.
value	MEDIUMTEXT	Cached value.
expiration	INT | INDEX	Expiration time as a Unix timestamp.

cache_locks

Field	Type / Key	Description
key	VARCHAR(255) | PK	Cache lock key.
owner	VARCHAR(255)	Lock owner.
expiration	INT | INDEX	Lock expiration time.

jobs		
Field	**Type / Key**	**Description**
id	BIGINT UNSIGNED | PK; AUTO_INCREMENT	Queued job identifier.
queue	VARCHAR(255) | INDEX	Queue name.
payload	LONGTEXT	Serialized job payload.
attempts	TINYINT UNSIGNED	Number of processing attempts.
reserved_at	INT UNSIGNED | NULL	Reservation time.
available_at	INT UNSIGNED	Time the job becomes available.
created_at	INT UNSIGNED	Job creation time.

job_batches		
Field	**Type / Key**	**Description**
id	VARCHAR(255) | PK	Batch identifier.
name	VARCHAR(255)	Batch name.
total_jobs	INT	Total jobs in batch.
pending_jobs	INT	Pending jobs.
failed_jobs	INT	Failed jobs.
failed_job_ids	LONGTEXT	Serialized failed job identifiers.
options	MEDIUMTEXT | NULL	Batch options.
cancelled_at	INT | NULL	Cancellation time.
created_at	INT	Batch creation time.
finished_at	INT | NULL	Completion time.

failed_jobs		
Field	**Type / Key**	**Description**
id	BIGINT UNSIGNED | PK; AUTO_INCREMENT	Failed job row identifier.
uuid	VARCHAR(255) | UQ	Unique failed job UUID.
connection	TEXT	Queue connection name.
queue	TEXT	Queue name.
payload	LONGTEXT	Serialized failed job payload.
exception	LONGTEXT	Failure exception details.
failed_at	TIMESTAMP | DEFAULT CURRENT_TIMESTAMP	Failure time.

personal_access_tokens		
Field	**Type / Key**	**Description**
id	BIGINT UNSIGNED | PK; AUTO_INCREMENT	Token row identifier.
tokenable_type	VARCHAR(255) | INDEX (morph pair)	Model type for token owner.
tokenable_id	BIGINT UNSIGNED | INDEX (morph pair)	Model identifier for token owner.
name	TEXT	Token name.
token	VARCHAR(64) | UQ	Unique token hash.
abilities	TEXT | NULL	Token permissions.
last_used_at	TIMESTAMP | NULL	Most recent token use.
expires_at	TIMESTAMP | NULL; INDEX	Token expiration time.
created_at	TIMESTAMP | NULL	Record creation time.
updated_at	TIMESTAMP | NULL	Most recent update time.